

2025 HKDSE Physics

Paper 1B — Structured Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Question 1

An empty glass cup is placed on a table under room temperature of $23\text{ }^{\circ}\text{C}$. 400 g of water at $100\text{ }^{\circ}\text{C}$ is poured into the cup. After a moment, the temperature of water in the cup is found to be $93\text{ }^{\circ}\text{C}$.

- (a) State a cause for the drop of water temperature other than heat loss to the surrounding air. (1 mark)
- (b) Estimate the heat capacity H , in $\text{J }^{\circ}\text{C}^{-1}$, of the glass cup. (2 marks)

Given: specific heat capacity of water = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- (c) (i) What is the kind of electromagnetic waves, other than visible light, mainly emitted from this cup of hot water? (1 mark)
- (ii) Name a device in daily life which makes use of this kind of electromagnetic waves. (1 mark)

Official marking scheme

Solution	Marks	Remarks
1. (a) The glass cup absorbs thermal energy (from the hot water) / Heat is lost to the glass cup.	1A	
	1	
(b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$ $H = 168\text{ (J }^{\circ}\text{C}^{-1}) \approx 170\text{ (J }^{\circ}\text{C}^{-1})$	1M 1A	
	2	
(c) (i) Infra-red (radiation) / thermal radiation	1A	
	1	
(ii) Remote control / infra-red (IR) thermometer / night vision camera	1A	
	1	

(a) Thermal energy is transferred from the hot water to the glass cup.

(b) $0.4 \times 4200 \times (100 - 93) = H(93 - 23)$, so $H = 168 \approx 170\text{ J }^{\circ}\text{C}^{-1}$.

(c) (i) Infrared radiation / thermal radiation.

(ii) Any valid use, e.g. a remote control, infrared thermometer or night-vision camera.

Question 2

A trolley is moving on a horizontal surface with a constant velocity of 2 m s^{-1} . At time $t = 0\text{ s}$, a small bead of mass 0.01 kg is projected upwards with a vertical speed of 5 m s^{-1} from a device P mounted on the trolley as shown. Neglect air resistance.



Figure 2.1

- *(a) Find the time taken, t , for the bead to travel back to the level of projection. (2 marks)
- (b) What is the kinetic energy of the bead at its maximum height? (2 marks)
- *(c) When the bead falls back to the level of projection, it is just above device P as shown. Explain why the bead does not fall behind or beyond device P at that moment. (2 marks)

Official marking scheme

2. (a) $s = ut + \frac{1}{2}at^2$ $0 = 5t + \frac{1}{2}(-9.81)t^2$ $t = 1.0194 \text{ s} \approx 1.02 \text{ s}$ or 0 s (rejected) (b) speed of the bead at its maximum height = 2 m s^{-1} $\text{K.E.} = \frac{1}{2} \times 0.01 \times 2^2$ $= 0.02 \text{ J}$ (c) The bead and the trolley have the same horizontal velocity and thus travel the same horizontal distance (within the same time interval t).	1M 1A 2 1M 1A 2 1A 1A 2	Or $v = u + at$, $0 = 5 + (-9.81)(t/2)$ $t = 1.02 \text{ s}$ Accept: $t = 1.00 \sim 1.02 \text{ s}$
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(a) $0 = 5t - \frac{1}{2}(9.81)t^2$, giving $t = 1.02 \text{ s}$ (reject $t = 0$).

(b) At the highest point, the horizontal speed is still 2 m s^{-1} : $K = \frac{1}{2}(0.01)(2^2) = 0.020 \text{ J}$.

(c) The bead and cart initially have the same horizontal velocity. With no horizontal force on the bead, their horizontal displacements in the same time are equal, so the bead remains vertically above P.

Question 3

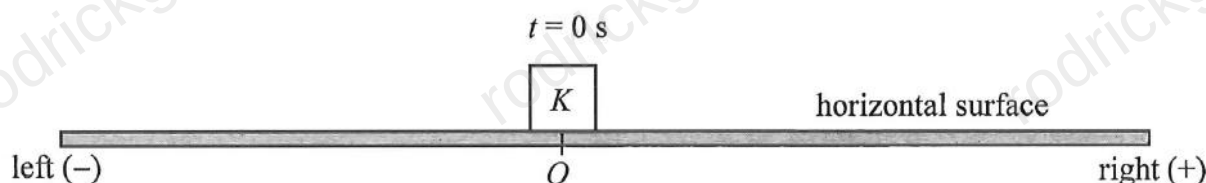
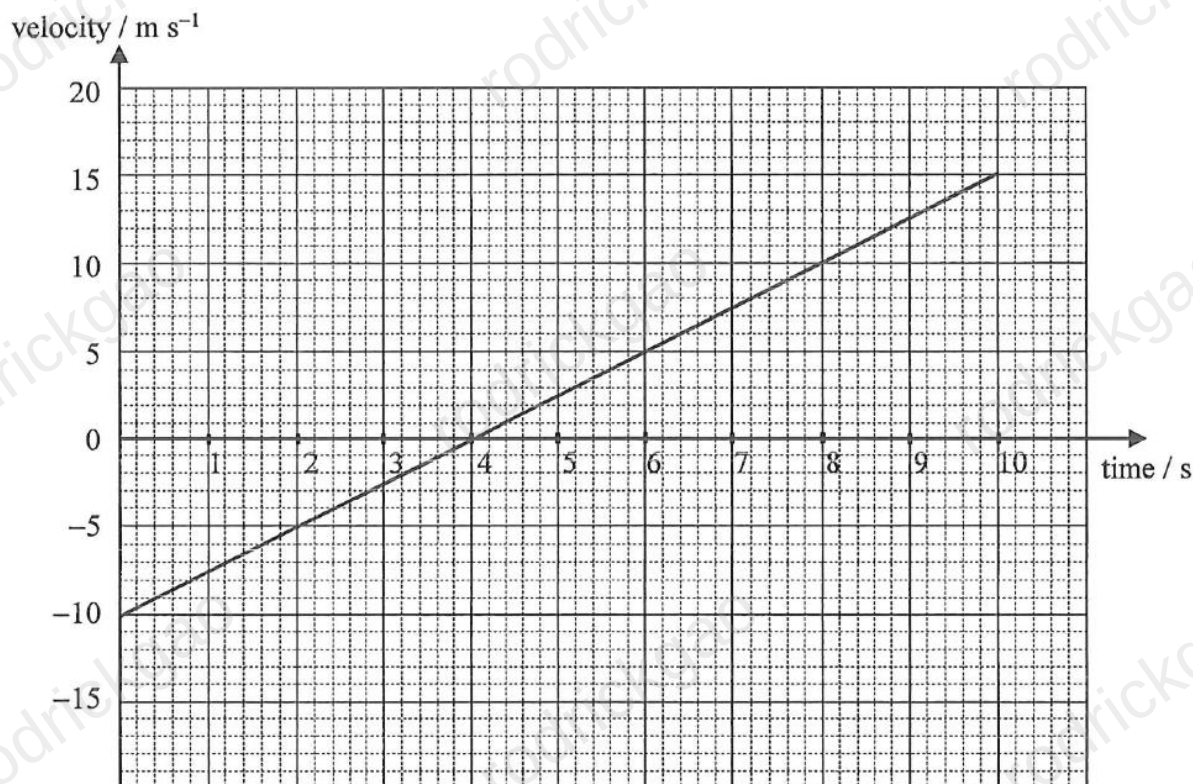


Figure 3.1

An object K moving on a horizontal surface is acted upon by a horizontal constant net force (not drawn on the diagram) such that it passes point O at time $t = 0 \text{ s}$ as shown in Figure 3.1. Its velocity–time graph is shown below, taking to the right as positive (+).



Velocity–time graph

- (a) Briefly describe the motion of the object from $t = 0$ s to $t = 4$ s. (2 marks)
- (b) Find the position of the object at $t = 10$ s. (2 marks)
- (c) Find the magnitude of the net force acting on the object at $t = 4$ s. The mass of the object is 0.4 kg. (2 marks)

Official marking scheme

Solution		
3. (a)	The object is moving to the left with constant deceleration and (momentarily) stops at $t = 4$ s.	1A 1A 2
(b)	Displacement = total area of the graph $= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$ $= 25 \text{ m}$ Position of the object is 25 m from point O on the right side.	1M 1A 2
(c)	Net force acting on the object $= ma$ $= 0.4 \times \frac{15 - (-10)}{10 - 0}$ $= 1 \text{ N (towards right)}$	1M 1A 2

(a) From 0–4 s, the object moves left and decelerates uniformly; it is momentarily at rest at $t = 4$ s.

(b) Displacement = signed area = $-\frac{1}{2}(10)(4) + \frac{1}{2}(15)(6) = +25$ m, so it is 25 m to the right of O.

(c) $a = [15 - (-10)]/10 = 2.5 \text{ m s}^{-2}$; $F = ma = 0.4 \times 2.5 = 1.0 \text{ N}$ to the right.

Question 4

A large clock facing upwards is placed horizontally on a table as shown. A small piece of plasticine Q rests on the steadily rotating secondhand of the clock.

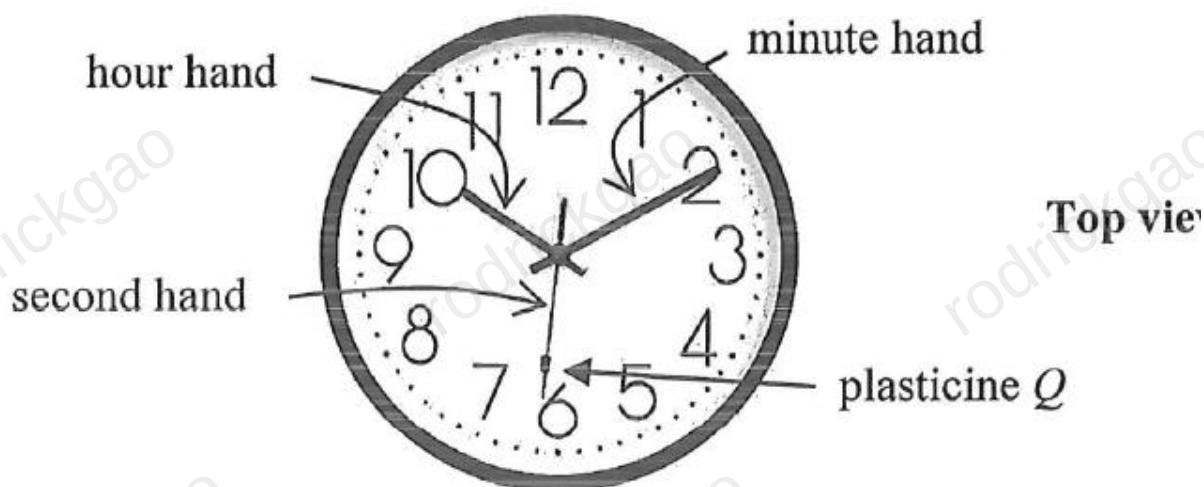


Figure 4.1

- (a) Find the angular speed ω , in rad s^{-1} , of the second hand of the clock. (1 mark)
- (b) Figure 4.2 shows the side view of the second hand of the clock.

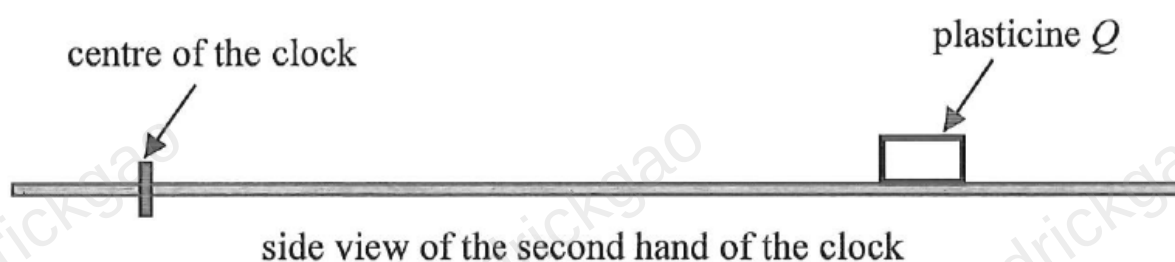


Figure 4.2

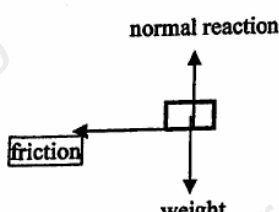
In the free-body diagram of plasticine Q below, add a force acting on Q to complete the diagram and name this force. (1 mark)



Free-body diagram

- (c) A student claims that the centripetal force required for Q will become smaller if it is placed nearer to the centre of the clock. Explain briefly whether this claim is correct or not. (2 marks)

Official marking scheme

4. (a) $\omega = \frac{2\pi}{60}$ $= \frac{\pi}{30}$ or $0.105 \text{ (rad s}^{-1}\text{)}$ (b)  (c) The required centripetal force $F_c = m\omega^2 r$, where r is the distance of plasticine Q from the clock's centre. As m and ω are constant, smaller r gives smaller centripetal force F_c. The claim is correct.	1A 1 1A 1 1A+1A 2	Accept: adhesive force
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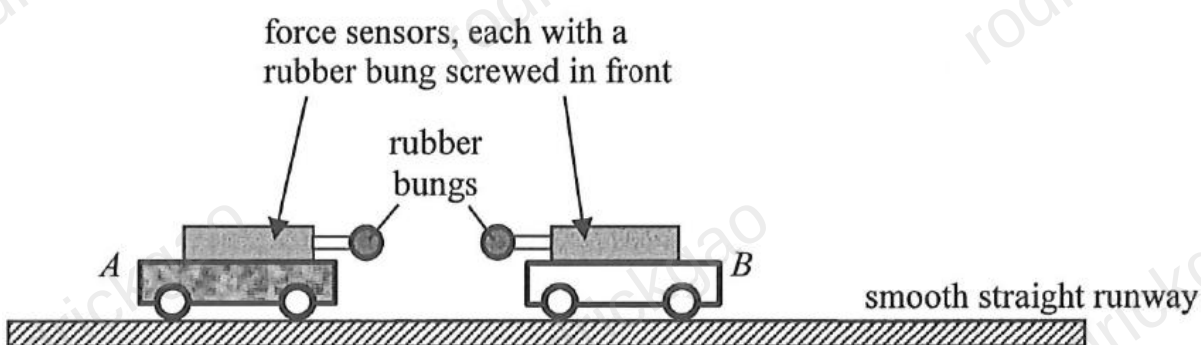
(a) $\omega = 2\pi/60 = \pi/30 = 0.105 \text{ rad s}^{-1}$.

(b) Add a frictional / adhesive force directed towards the centre of the clock.

(c) $F_c = m\omega^2 r$. Since m and ω are constant, decreasing r reduces the required centripetal force; the statement is correct.

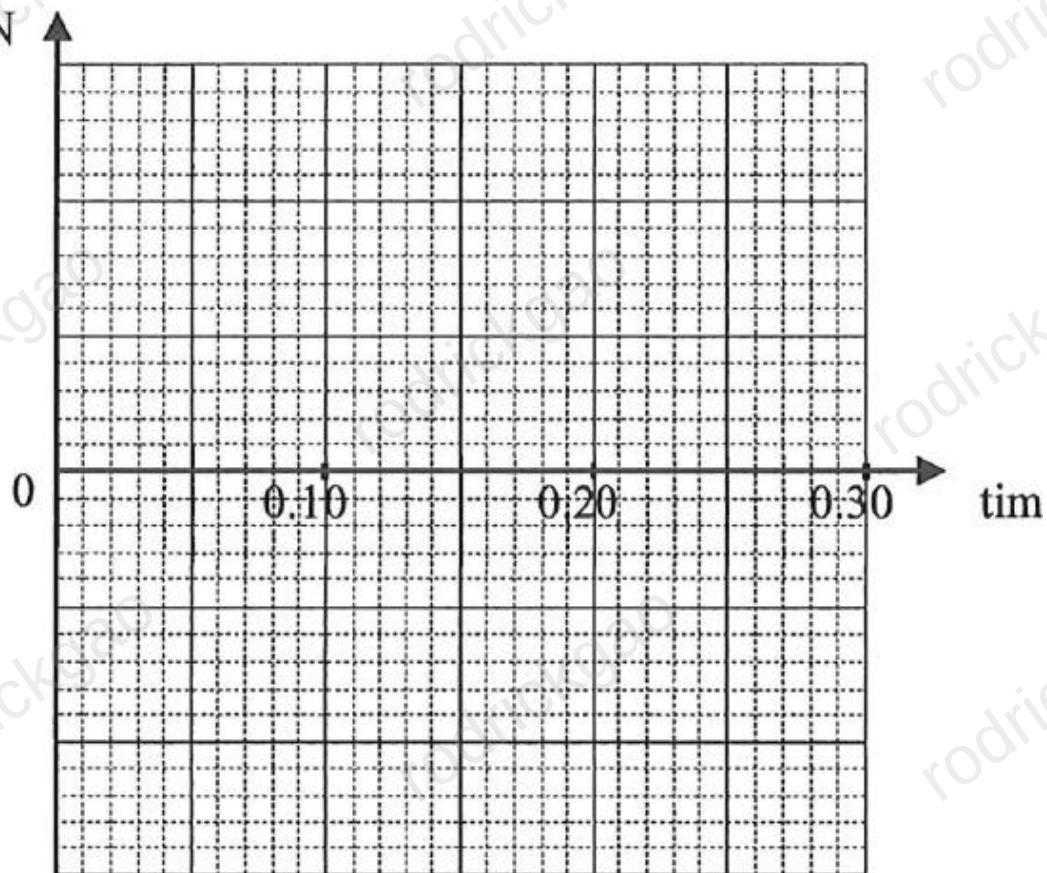
Question 5

You are given two trolleys A and B (with mass of trolley A being larger), which can run freely on a smooth horizontal runway. Each trolley has a force sensor mounted on it and the two sensors are connected to a computer interface to record the time variation of force (force towards the right is registered as positive).

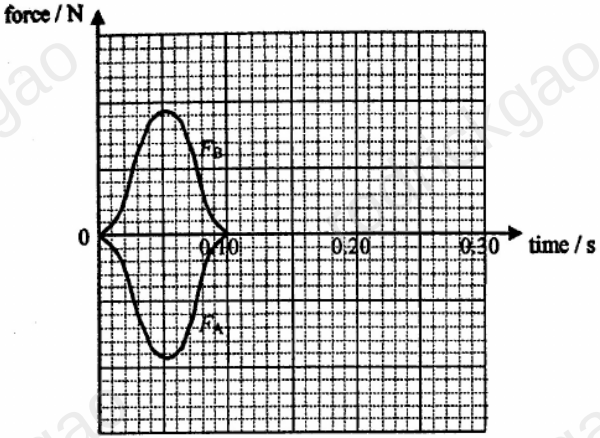


Collision set-up

A student claims that when two objects collide, the less massive object experiences a larger force of collision. Describe how you would conduct an experiment to investigate the forces acting between the trolleys during collision. Sketch the expected force–time graphs of the trolleys in the grid provided and explain briefly. Assume that the collision takes place at time 0 s and lasts for approximately 0.1 s. (4 marks)

*Force–time axes*

Official marking scheme

SOLUTION	MARKS	REMARKS
5. With the computer interface switched on for data recording (force towards the right taken as positive), give trolley <i>A</i> a push and let it collide head-on with trolley <i>B</i>. Force-time graphs:  The force acting on (the less massive) trolley <i>B</i>, i.e. F_B, is always equal (in magnitude and opposite in direction) to that acting on trolley <i>A</i>, i.e. F_A, as they are an action and reaction pair / according to Newton's 3rd law.	1A 2A 1A 4	

Start data logging and let cart *A* collide head-on with cart *B*. The two *F*–*t* curves should occur over the same ≈ 0 – 0.1 s interval and have equal magnitudes but lie on opposite sides of the zero line. By Newton's third law, the forces are equal and opposite at every instant; the lighter cart does not experience a larger force.

Question 6

A student builds a smartphone projector as shown in Figure 6.1. It consists of a paper box with a lens *L* in front and a movable smartphone support inside. The position of the smartphone can be adjusted until a sharp image is formed on a wall.

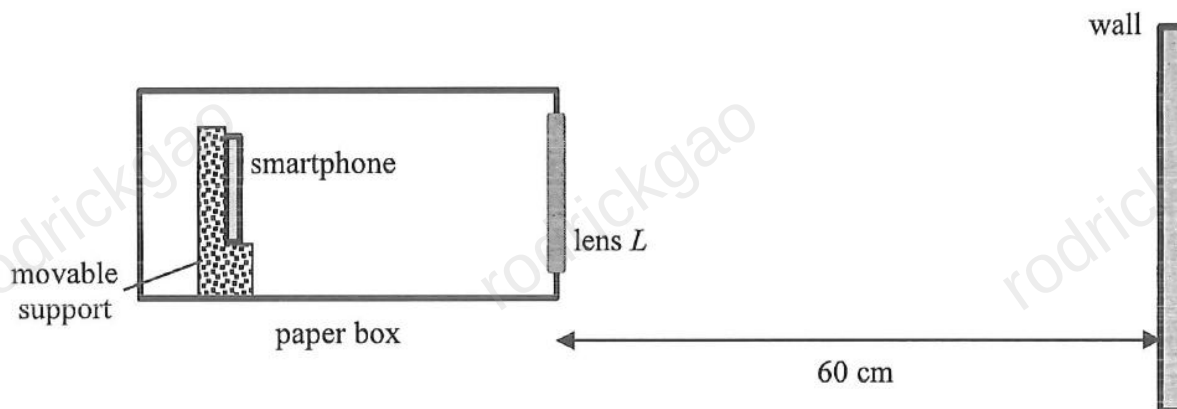
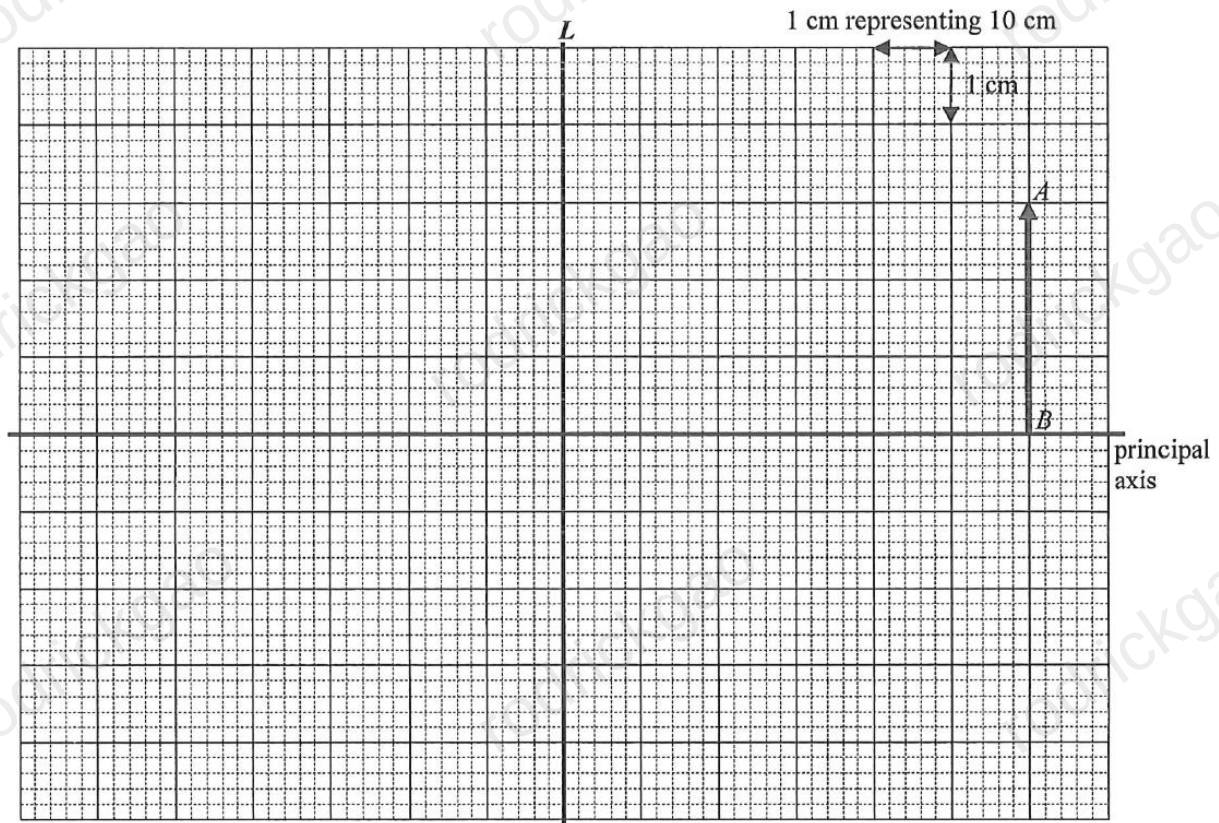


Figure 6.1

(a) What kind of lens is used? Explain briefly. (2 marks)

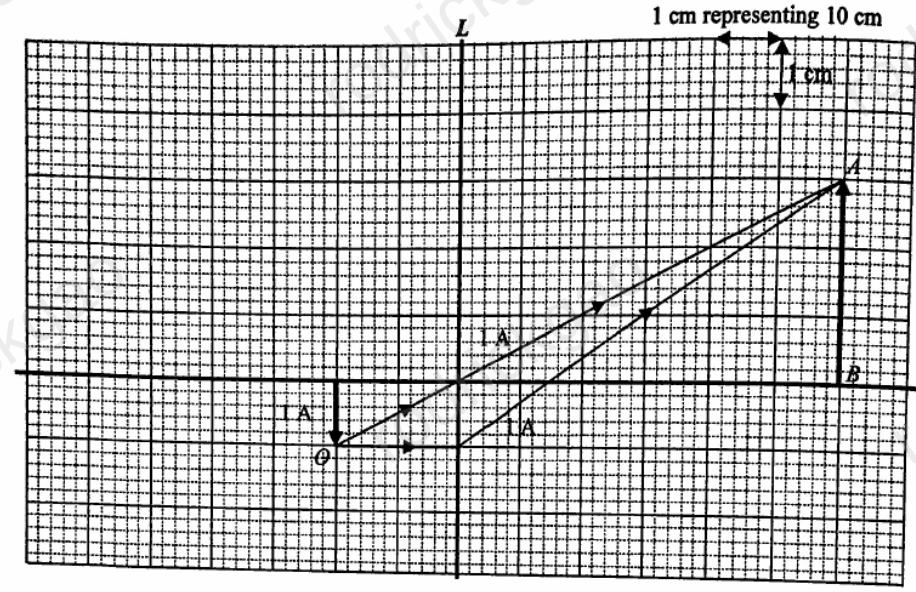
- (b) In the figure below, AB represents the image formed on the wall, which is 60 cm from lens L. Given that the magnification of the image is 3.



Ray-diagram grid

- Locate and draw the object (i.e. the display on the smartphone) using an arrow and denote it as O. (1 mark)
- Complete the ray diagram by drawing TWO light rays reaching A from the object. (2 marks)
- Hence, or otherwise, find the focal length of L. (1 mark)
- If the separation between lens L and the wall is slightly increased, state the adjustment to be made so as to restore a sharp image on the wall. (1 mark)
- Suggest a way to minimise unwanted reflections inside the box. (1 mark)

Official marking scheme

Solution	Marks	Remarks
6. (a) Convex lens because only a convex lens can form a real image.	1A 1A 2	Or the image can be captured on a screen / the image is formed on the opposite side of the lens / only convex lens can form a magnified image.
(b) (i)(ii) 		
(iii) Ray diagram method: $f = 15 \text{ cm}$	3 1A 1 1A 1A 1	Accept: 14 cm ~ 16 cm Or By $\frac{v}{u} = 3$, $\frac{0.6}{u} = 3$, i.e. $u = 0.2 \text{ m}$ By $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $\frac{1}{f} = \frac{1}{0.2} + \frac{1}{0.6}$ $f = 0.15 \text{ m or } 15 \text{ cm}$
(c) Move / adjust the support / smartphone closer to lens L.		
(d) The box should be painted black inside.		

(a) A convex lens, because only it can form a real image on the wall.

(b) Magnification $v/u = 3$ and $v = 60 \text{ cm}$, so $u = 20 \text{ cm}$. Draw the object inverted with one-third the image height; use a ray through the optical centre and a ray parallel to the axis then through the focus. $1/f = 1/20 + 1/60$, so $f = 15 \text{ cm}$.

(c) If the wall is moved slightly farther away, move the phone closer to the lens.

(d) Paint the inside black to absorb stray light.

Question 7

The circuit shown in Figure 7.1 can be used to measure the current at different lengths of a resistance wire using a movable jockey. The power supply of negligible internal resistance delivers a voltage of 3.5 V.

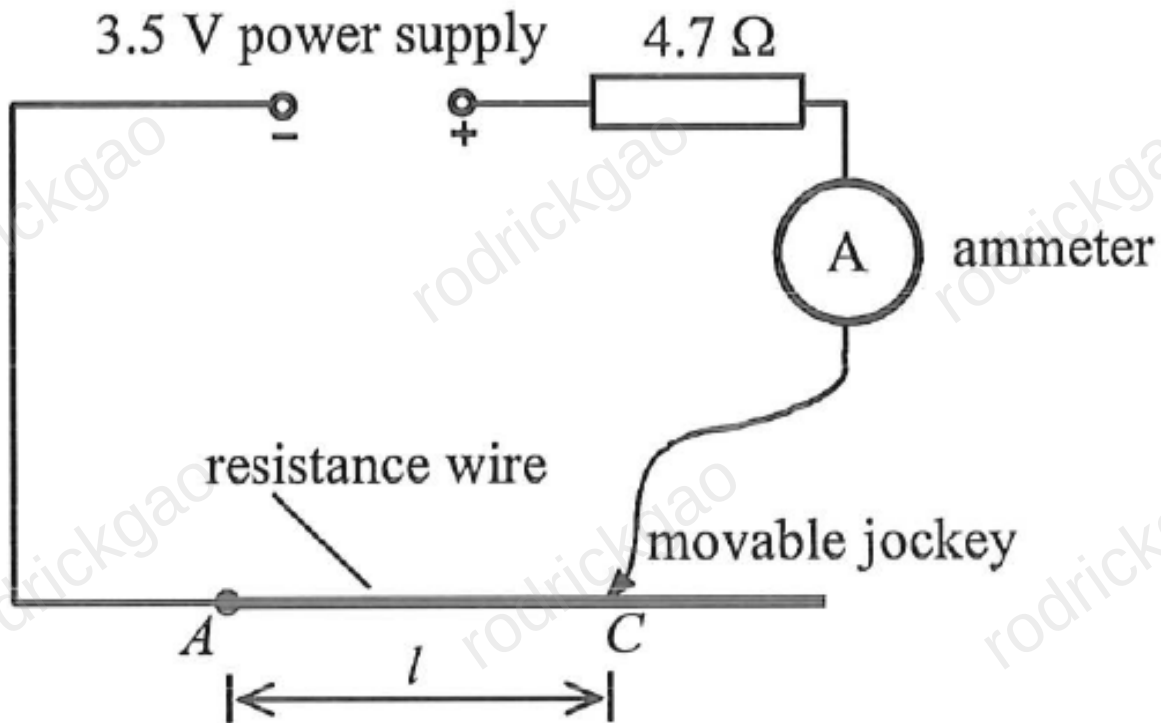
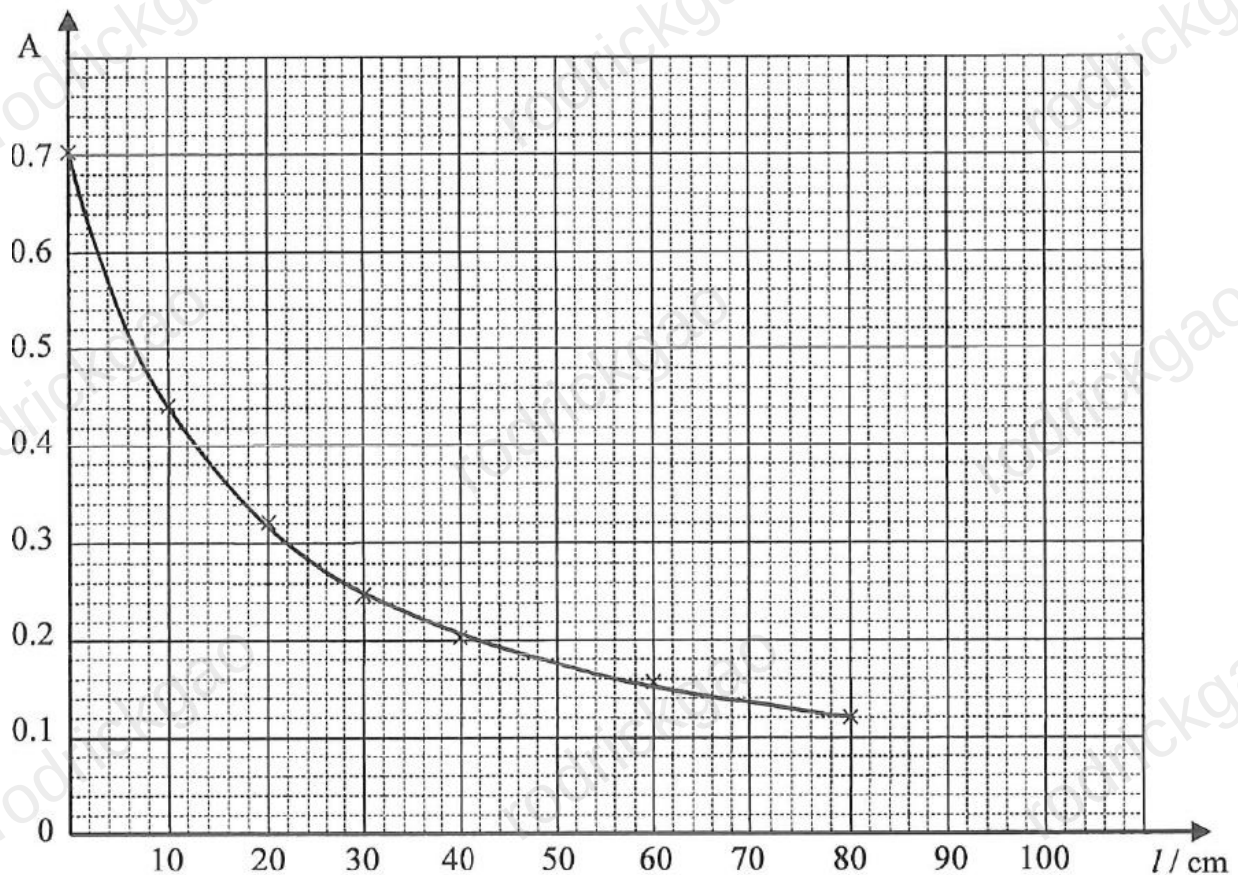


Figure 7.1

When the jockey is placed at point C on the resistance wire, the length of wire used is l and the corresponding current I is measured with an ammeter. The experiment is repeated at different lengths l and a graph of I against l is plotted below.



Current-length graph

- (a) What is the function of the $4.7\ \Omega$ resistor in this experiment? (1 mark)
- (b) When the jockey is placed at point A (i.e. $\ell = 0$), find the resistance of the whole circuit. Compare it with the value of the $4.7\ \Omega$ resistor and account for the difference. (3 marks)
- (c) When the jockey is moved from point A on the resistance wire to point C where $\ell = 30\text{ cm}$, find the increase in resistance of the circuit. (2 marks)

Official marking scheme

Solution	Marks	Remarks
7. (a) To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat.	1A	
	1	
(b) $R = \frac{V}{I} = \frac{3.5}{0.7}$ $= 5.0 \Omega (> 4.7 \Omega)$ The extra resistance (0.3Ω) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply.	1M 1A 1A	
	3	
(c) $I = 0.25 \text{ A}$ at $\ell = 30 \text{ cm}$ $\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$ $= 9.0 \Omega$	1M 1A	Accept: $I = 0.24 \text{ A} \sim 0.26 \text{ A}$ e.c.f. from (b) Accept: $\Delta R = 8.5 \Omega \sim 9.6 \Omega$
	2	

(a) It limits current, protecting the ammeter / preventing overheating.

(b) At $\ell = 0$, the graph gives $I \approx 0.70 \text{ A}$, so $R = 3.5/0.70 = 5.0 \Omega$. The extra 0.3Ω over 4.7Ω comes from the ammeter, contacts or source internal resistance.

(c) At $\ell = 30 \text{ cm}$, $I \approx 0.25 \text{ A}$; $\Delta R = 3.5/0.25 - 5.0 = 9.0 \Omega$.

Question 8

A rectangular coil, with N turns and each turn of area $9.00 \times 10^{-4} \text{ m}^2$, is placed in a uniform magnetic field of strength B . Initially the plane containing the coil is parallel to the magnetic field. The coil is then rotated steadily about the axis XY as shown in Figure 8.1(a).

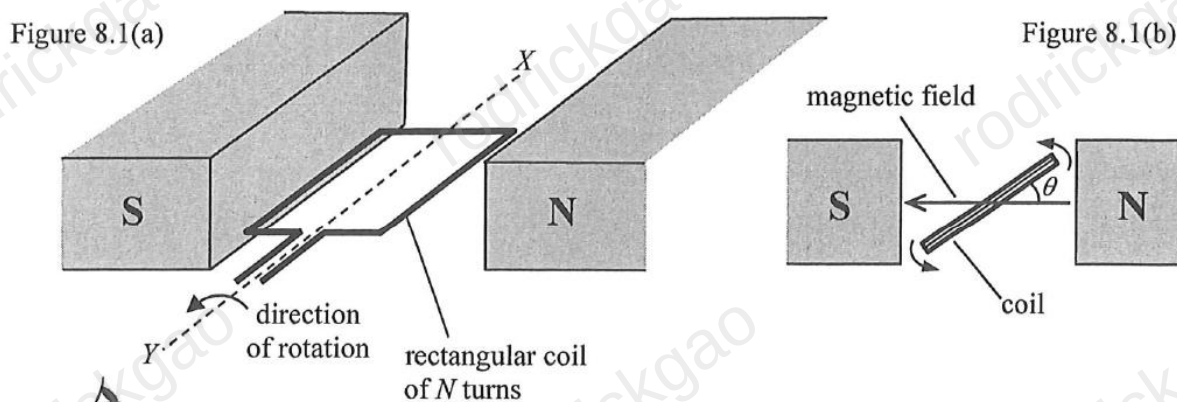
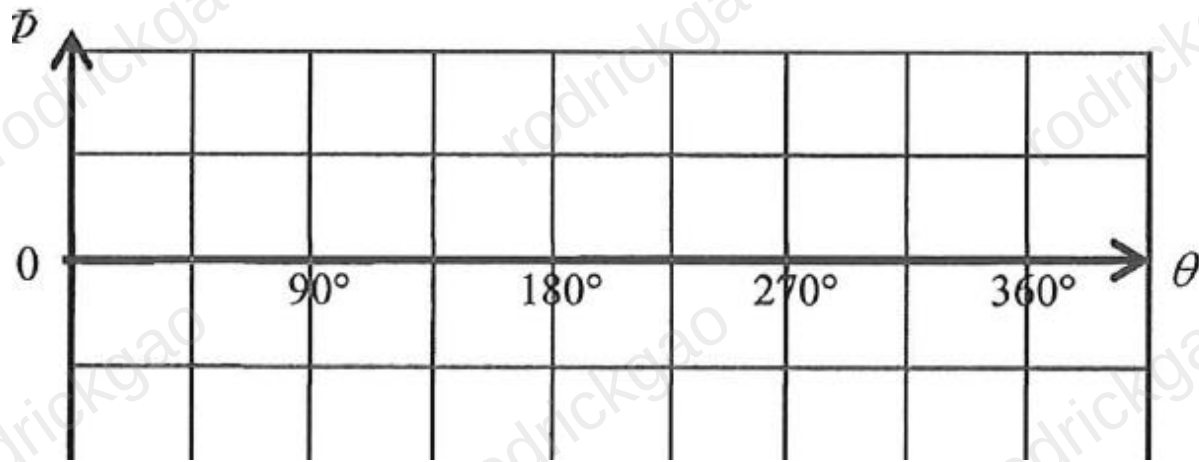


Figure 8.1

- (a) Sketch the variation of magnetic flux Φ passing through each turn of the coil when the coil rotates from $\theta = 0^\circ$ to 360° (see Figure 8.1(b)). (2 marks)



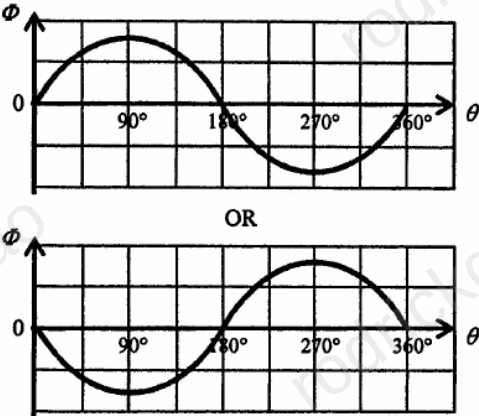
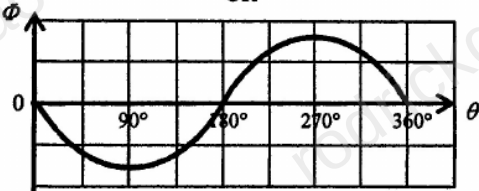
Magnetic-flux graph

(b) The maximum magnetic flux through each turn of the coil is 2.70×10^{-4} Wb.

(i) Determine B (unit: T). (2 marks)

(ii) Calculate the average induced e.m.f. ε in the coil when it is rotated from $\theta = 0^\circ$ to 90° in 0.01 s. Given that the number of turns N is 100. (2 marks)

Official marking scheme

8. (a)  OR 	2A	
(b) (i) $\phi = BA \cos 0^\circ = 2.70 \times 10^{-4}$ $B \times (9.00 \times 10^{-4}) \times 1 = 2.70 \times 10^{-4}$ $B = 0.300 \text{ (T)}$	2	
(ii) $\varepsilon = \left -N \frac{\Delta \phi}{\Delta t} \right$ $= \left -100 \times \frac{2.70 \times 10^{-4}}{0.01} \right$ $= 2.70 \text{ V}$	1M 1A 2	Accept: -2.70 V

(a) Φ varies sinusoidally with θ : zero at 0° , maximum at 90° , zero at 180° , opposite maximum at 270° , and zero at 360° .

(b) (i) $B = \Phi_{\max}/A = (2.70 \times 10^{-4})/(9.00 \times 10^{-4}) = 0.300 \text{ T}$.

(ii) $|\varepsilon_{\text{avg}}| = N|\Delta\Phi|/\Delta t = 100(2.70 \times 10^{-4})/0.01 = 2.70 \text{ V}$.

Question 9

An ideal gas is at temperature 0°C and pressure $1.01 \times 10^5 \text{ N m}^{-2}$.

(a) *(i) Find the volume of one mole of the gas. (2 marks)

(ii) Find the number of molecules in 1 m^3 of the gas under the conditions given. (2 marks)

*(b) At a certain location above the Earth in space where the pressure is $2.75 \times 10^{-10} \text{ N m}^{-2}$, 1 m^3 of the gas contains about 1.00×10^{11} molecules. Assuming the gas is ideal, estimate the temperature at the location (in K). (2 marks)

Official marking scheme

Solution	Marks	Remarks
9. (a) (i) $pV = nRT$ $1.01 \times 10^5 \times V = 1 \times 8.31 \times 273$ $V = 2.2462 \times 10^{-2} \text{ m}^3 \approx 2.25 \times 10^{-2} \text{ m}^3$	1M 1A 2	Accept: $0.0220 \sim 0.0225 \text{ m}^3$
(ii) Number of molecules in 1 m^3 $= \frac{N_A}{\text{volume of one mole of molecule}}$ $= \frac{6.02 \times 10^{23}}{2.246 \times 10^{-2}}$ $= 2.680120 \times 10^{25} \approx 2.68 \times 10^{25}$	1M 1A 2	e.c.f. from (a)(i) Or $pV = nRT$ $1.01 \times 10^5 \times 1 = n \times 8.31 \times 273$ $n = 44.5$ $N = 44.5 \times 6.02 \times 10^{23}$ $= 2.68 \times 10^{25}$
(b) $pV = \left(\frac{N}{N_A} \right) RT$ $2.75 \times 10^{-2} \times 1 = \left(\frac{1.00 \times 10^{11}}{6.02 \times 10^{23}} \right) \times 8.31 \times T$ $T = 199.2178 \text{ (K)} \approx 199 \text{ (K)}$	1M 1A 2	Accept: $199 \sim 200 \text{ K}$

(a) (i) $V = nRT/p = (1)(8.31)(273)/(1.01 \times 10^5) = 2.25 \times 10^{-2} \text{ m}^3$.

(ii) $N/V = N_A/V_m = 6.02 \times 10^{23}/(2.246 \times 10^{-2}) = 2.68 \times 10^{25} \text{ m}^{-3}$.

(b) From $pV = (N/N_A)RT$, $T = pVN_A/(NR) = 199 \text{ K}$.

Question 10

- (a) A student uses the set-up in Figure 10.1 to investigate stationary waves on a stretched string. When the frequency of the vibrator is set at 100 Hz, a stationary wave is formed between the vibrator and the stand, which are 90 cm apart.

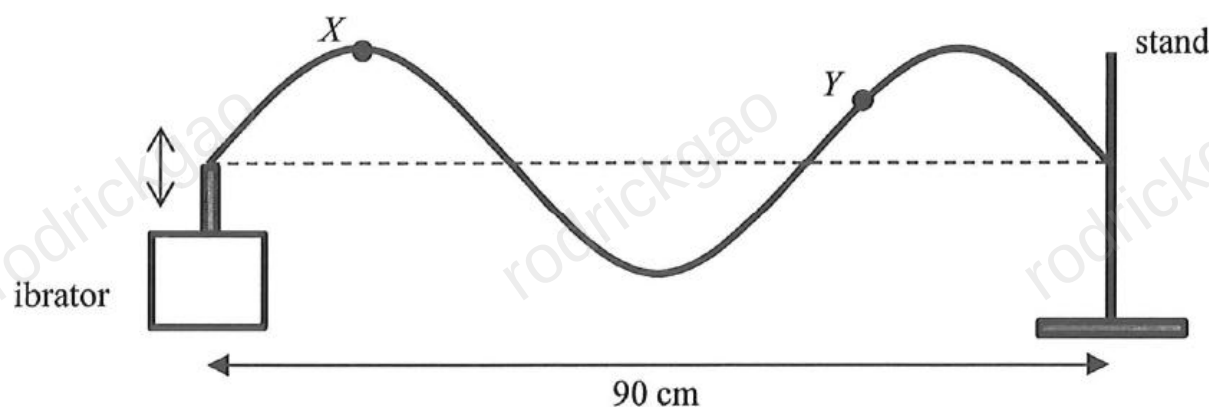


Figure 10.1

Particles X and Y on the string are at maximum displacement from their respective equilibrium positions at the moment shown in Figure 10.1.

- (i) Label the positions of all antinodes (using 'A') and nodes (using 'N') on Figure 10.1. (2 marks)
 - (ii) Compare the oscillations of X and Y in terms of their amplitude and phase. (2 marks)
 - (iii) Sketch on Figure 10.1 the shape of the string just after 0.0075 s. (1 mark)
 - (iv) If the frequency of the vibrator is increased gradually, at what frequency will the next stationary wave pattern be observed? Show your working. (2 marks)
- (b) Read the following passage about stationary waves in musical instruments and answer the questions that follow.

How do we make musical sounds? In non-electronic instruments such as guitar and piano, stable and controlled vibration is usually produced by a stationary wave. If you pluck a guitar string fixed at both ends, various modes of stationary waves will be produced and they all travel with the same speed on the string. The mode with the lowest frequency f_1 is called the fundamental while the n th mode, called the n th harmonic, has a frequency n times that of the fundamental, i.e. nf_1 .

The pitch of a note is determined by the frequency of the fundamental. The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments. Therefore, a guitar and a violin playing a note of the same pitch will have different qualities because of different amplitudes and distributions of their harmonics.

- (i) Figure 10.2 shows the stationary wave pattern of the fundamental formed on a string. Sketch the stationary wave pattern corresponding to the 2nd harmonic on Figure 10.2. (1 mark)

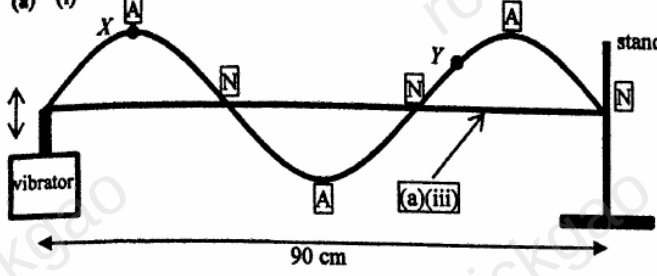
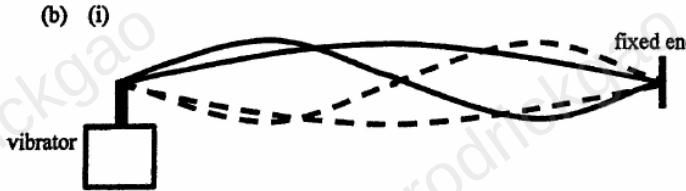


Figure 10.2

vibrator

- (ii) Explain why different musical instruments give different sound qualities. (2 marks)
- (iii) State TWO main differences between the waves on a vibrating guitar string and the sound waves produced by it besides the two waves being different in wavelength and speed. (2 marks)

Official marking scheme

Solution	Marks	Remarks
10. (a) (i)  (ii) Particle X vibrates with a larger amplitude than particle Y while they are vibrating in phase. (iii) Correct sketch of a straight string (iv) Original wavelength $\lambda_1 = \frac{0.90}{1.5} = 0.60 \text{ m}$ at $f_1 = 100 \text{ Hz}$ For the next stationary wave pattern, $\lambda_2 = \frac{0.90}{2} = 0.45 \text{ m}$ By $v = f\lambda$, $100 \times 0.60 = f_2 \times 0.45$ $f_2 = 133.33 \text{ Hz} \approx 130 \text{ Hz}$	2A 2 1A 1A 2 1A 1 1M 1A 2	Or $f_2 = 100 \times \frac{4}{3}$ $= 133.33 \text{ Hz} \approx 130 \text{ Hz}$
(b) (i)  (ii) The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments, when these harmonics superimpose and combine to produce different waveforms. (iii) Any TWO: stationary vs travelling transverse vs longitudinal energy confined in a string vs energy being transferred through air	1A 1 1A 1A 2 2A 2	

(a) (i) In 90 cm there are 3 antinodes and 4 nodes, including both ends and two internal nodes.

(ii) X has a larger amplitude than Y; they are in phase.

(iii) $0.0075 \text{ s} = 3T/4$, so the string is momentarily straight.

(iv) $\lambda = 0.90/1.5 = 0.60 \text{ m}$ and wave speed $= 100 \times 0.60 = 60 \text{ m s}^{-1}$. For the next mode, $\lambda = 0.90/2 = 0.45 \text{ m}$, so $f = 60/0.45 = 133 \approx 130 \text{ Hz}$.

(b) (i) Draw the second harmonic with two antinodes and a central node.

- (ii) Different relative harmonic amplitudes produce different resultant waveforms and hence different timbres.
- (iii) Any two contrasts: standing/travelling, transverse/longitudinal, or energy retained on the string/transferred through air.

Question 11

A coal-fired power plant consumes 4.5×10^6 kg of coal per day to generate electric power at an average rate of 680 MW. Given that burning 1 kg of coal releases 32 MJ of energy.

- (a) (i) Find the daily energy input (in MJ). (2 marks)
- (ii) Find the electrical energy produced per day (in MJ). (2 marks)
- (iii) Estimate the efficiency of this power plant. (2 marks)
- (b) Transmission cables at 400 kV are used for long-distance electric power transmission to a city.
 - (i) Estimate the current in the cables. (2 marks)
 - (ii) Estimate the power loss (in kW) in the transmission cables. The resistance of the cables is $2.90 \times 10^{-8} \Omega$ per metre and the total length of the cables is 160 km. (2 marks)
 - *(iii) Explain why a.c. is preferred for electric power transmission. (2 marks)

Official marking scheme

Solution		
11. (a) (i) $E_{in} = (32 \times 10^6) \times (4.5 \times 10^6)$ $= 1.44 \times 10^{14} \text{ J} \approx 1.4 \times 10^8 \text{ (MJ)}$	1M	
	1A	
	2	
(ii) $E_{out} = (680 \times 10^6) \times (24 \times 3600)$ $= 5.875 \times 10^{13} \text{ J} \approx 5.9 \times 10^7 \text{ (MJ)}$	1M	
	1A	
	2	
(iii) Efficiency = $\frac{E_{out}}{E_{in}} \times 100\%$ $= \frac{5.88 \times 10^{13}}{1.44 \times 10^{14}} \times 100\%$ $= 40.8\% \approx 41\%$	1M	e.c.f. from (a)(i)(ii) Accept: 41% ~ 42%
	1A	
	2	
(b) (i) $P = VI$ $680 \text{ MW} = 400 \text{ kV} \times I$ $I = 1700 \text{ A} = 1.7 \text{ kA}$	1M	
	1A	
	2	
(ii) Power loss = $I^2 R$ $= (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3)$ $= 13410 \text{ W} \approx 13.4 \text{ (kW)}$	1M	e.c.f. from (b)(i)
	1A	
	2	
(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective. High voltage transmission has low transmission current and so reduces power loss significantly.	1A	
	1A	
	2	

(a) (i) $E_{in} = (4.5 \times 10^6)(32 \times 10^6) = 1.44 \times 10^{14} \text{ J} = 1.44 \times 10^8 \text{ MJ}$.

(ii) $E_{out} = (680 \times 10^6)(24 \times 3600) = 5.88 \times 10^{13} \text{ J} = 5.88 \times 10^7 \text{ MJ}$.

(iii) $\eta = E_{out}/E_{in} \times 100\% = 40.8\% \approx 41\%$.

(b) (i) $I = P/V = 680 \text{ MW}/400 \text{ kV} = 1.70 \text{ kA}$.

(ii) $R = (2.90 \times 10^{-8})(160 \times 10^3) = 4.64 \times 10^{-3} \Omega$; $P_{loss} = I^2 R = 13.4 \text{ kW}$.

(iii) A.C. can be stepped up by a transformer; for the same power, higher V gives smaller I and therefore much smaller $I^2 R$ loss.

Question 12

Carbon-14 (^{14}C) nuclide decays into stable nitrogen-14 (^{14}N) nuclide with a half-life of 5730 years. Assume that the concentration of carbon-14 in the atmosphere remains unchanged over time.

(a) (i) Write an equation to represent the decay of carbon-14. (1 mark)

*(ii) Find the decay constant (in yr^{-1}) of carbon-14. (2 marks)

*(iii) 1 g of an ancient wood sample has a carbon-14 activity that gives a count rate of 6.0 counts per minute. Estimate the age of this sample in years. Given that the carbon-14 activity in 1 g of fresh wood gives a count rate of 13.6 counts per minute. (2 marks)

- (b) A more advanced method involves directly measuring the number of carbon-14 atoms relative to that of the carbon-12 atoms. Carbon atoms from the wood sample are ionized to form a beam which is directed into a magnetic field as shown in Figure 12.1. Each of the positive carbon-14 and carbon-12 ions possesses one electronic charge in magnitude.

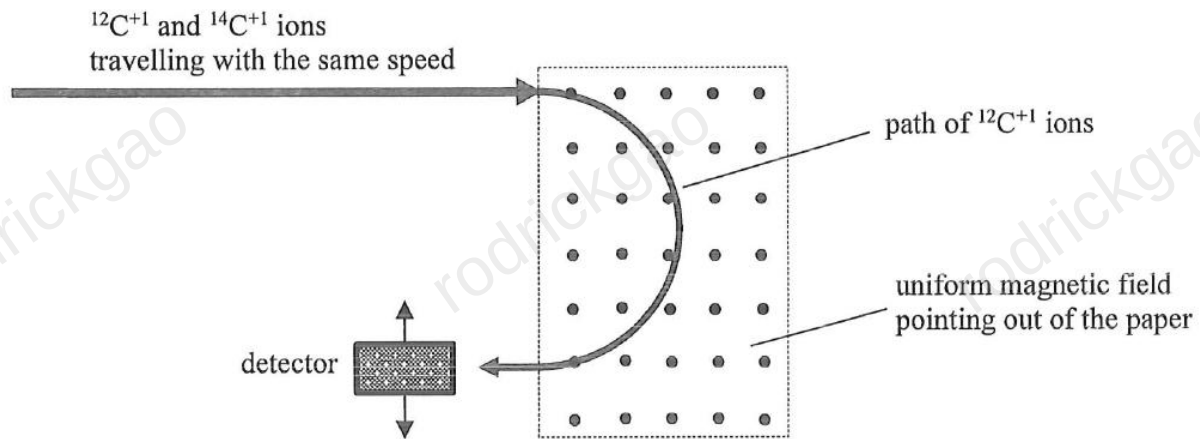
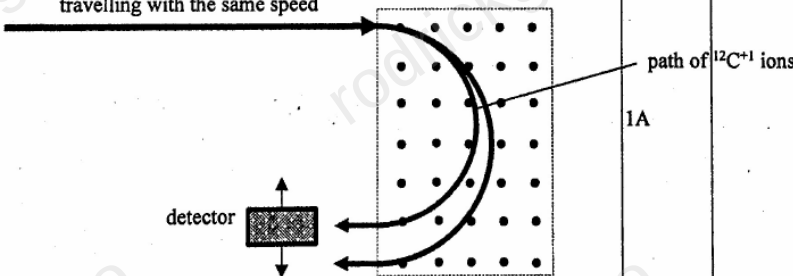


Figure 12.1

- (i) Both carbon-14 and carbon-12 are isotopes of carbon. What is meant by 'isotopes'? (1 mark)
- *(ii) Sketch the path of $^{14}\text{C}^{+}$ ions on Figure 12.1. (1 mark)
- (iii) Explain why the speed of the ions remains unchanged after passing through the magnetic field. (2 marks)

Official marking scheme

Solution	Marks	Remarks
12. (a) (i) ${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{N} + {}^0_{-1}\text{e}$ (or β^-)	1A	
	1	
(ii) $k = \frac{\ln 2}{t_{\frac{1}{2}}}$ $= \frac{\ln 2}{5730}$ $= 1.2097 \times 10^{-4} \text{ (yr}^{-1}) \approx 1.21 \times 10^{-4} \text{ (yr}^{-1})$	1M 1A	
	2	
(iii) $C = C_0 e^{-kt}$ $6.0 = 13.6 e^{-(1.21 \times 10^{-4})t}$ $t = 6764.7 \approx 6800 \text{ (years)}$	1M 1A	e.c.f. from (a)(ii) Accept: 6760 ~ 6820 (years)
	2	
(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.	1A	Or same number of protons but different number of neutrons
	1	
(ii) ${}^{12}\text{C}^{+1}$ and ${}^{14}\text{C}^{+1}$ ions travelling with the same speed		
	1A	
	1	
(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion, no work is done on the ions.	1A 1A	
	2	

(a) (i) Carbon-14 $\rightarrow$ nitrogen-14 + β^- .(ii) $\lambda = \ln 2 / 5730 = 1.21 \times 10^{-4} \text{ yr}^{-1}$.(iii) $6.0 = 13.6 e^{-\lambda t}$, so $t = \ln(13.6/6.0) / \lambda = 6.76 \times 10^3 \approx 6800 \text{ yr}$.

(b) (i) Isotopes have the same proton number / atomic number but different neutron numbers / mass numbers.

(ii) A positive ion bends downward in the field out of the page. Since $r = mv / (qB)$, carbon-14 has the larger radius and its path is drawn outside that of carbon-12.

(iii) Magnetic force is always perpendicular to velocity and does no work, so kinetic energy and speed remain unchanged.